Mahmudul Haque

Software Engineer | C++, Computer Vision & AI

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Professional Profile

MSc Automotive Software Engineering student at TU Chemnitz, based in Chemnitz, with professional experience in C++, computer vision, machine learning, hardware-oriented desktop software, backend development, and requirements engineering. Conducted semiconductor R&D involving 3D height-data image stitching, OpenCV pattern recognition, wafer mapping, and position-based autofocus compensation. Strong in algorithms, object-oriented and modular design, and technical problem-solving, supported by 700+ solved programming problems and two first-place contest finishes. Highly comfortable with Linux/Unix systems.

Technical Skills

AI Productivity: Strong proficiency in prompt engineering and effective use of AI agents
Productivity: Microsoft Office
Web Project Toolkit: TypeScript, PHP, Next.js, Node.js, Express, Tailwind CSS, PostgreSQL, Prisma, Three.js
Hardware/Lab: Arduino, Raspberry Pi, PM/humidity sensors, digital displays, university prototyping
Programming: C++, C, C#, Python, Java, JavaScript, HTML, CSS
AI/Computer Vision: Computer vision, image stitching, geometric mapping, pattern recognition, LLMs, generative AI, deep learning, CNN, RNN, Deep Q-Networks
AI/ML Libraries: OpenCV, MediaPipe, DeepFace, RetinaFace, PyTorch, TensorFlow, CUDA, scikit-learn
LLM/Web/Backend: Local LLMs, RAG, embeddings, agentic workflows, FastAPI, LangChain, ChromaDB, Ollama, LM Studio, React, ASP.NET Core, REST APIs, server-sent events
Cross-Platform/Systems: Qt, Linux/Unix, Fedora (daily use), Linux CLI, Windows, Docker, SSH
Desktop/Engineering: MFC, Visual Studio, GDI+, Java Swing, OOP, data structures, algorithms, modular design, code refactoring, software architecture
Build/Tools: Git, Visual Studio solutions/projects, .NET CLI, Docker Compose, Postman
Data/Cloud: SQL, PySpark, SQL Server, MySQL, Firebase, Apache Derby, AWS EC2, AWS S3
Simulation/Graphics: SUMO, OpenGL, iGraphics
Current interests: Cross-platform C++, embedded systems, CMake, multimodal AI, real-time interaction.

Professional Experience

Software Engineer (R&D) *Nov 2025 – Present*
Frontier Semiconductor Metrology (FSM), Dhaka, Bangladesh

- Built object-oriented C++/MFC software for tool control, automation, data processing, and visualization.
- Worked on CUDA-based image processing for semiconductor metrology equipment.
- Stitched multiple 3D height-data images into one continuous die image when the camera field of view could not cover the whole die.
- Worked on OpenCV pattern recognition within wafer dies for structure inspection.
- Reconstructed a full virtual wafer die map from die coordinates calibrated using four points across three adjacent dies, enabling navigation to specific dies for inspection in C++/MFC.
- Fixed stage-tilt autofocus issues using a virtual plane to adjust Z-axis focus height based on the probe's wafer position.

Technical Business Analyst *Aug 2025 – Oct 2025*
US-Bangla Airlines, Dhaka, Bangladesh

- Mapped FlightOps requirements with cabin/cockpit crews, scheduling executives, and the Operations Control Center.
- Converted stakeholder needs into technical specifications, software documentation, and developer UI/UX wireframes.
- Designed crew-training backend rules for training cycles, certificate expiry, and programme assignments.

Education

MSc in Automotive Software Engineering *Oct 2026 – Present*
Chemnitz University of Technology (TU Chemnitz), Chemnitz, Germany

BSc in Computer Science and Engineering *Jun 2021 – Jun 2025*
Ahsanullah University of Science and Technology, Dhaka, Bangladesh
CGPA: 3.719/4.00 **German equivalent:** 1.4 Degree record

Projects

HCFD Community Platform Website
Next.js, TypeScript, PostgreSQL, Prisma

- Built a bilingual Bengali/English community website and administration dashboard.

Growthy Solutions Website Website
Next.js, React, TypeScript

- Developed a responsive studio website with service information, accessible navigation, and contact interactions.

Live Weather-Index Display – University Lab
Arduino, Raspberry Pi, environmental sensors, digital display

- Measured particulate matter and humidity and passed sensor readings to a model to predict a weather index.
- Displayed the predicted index on a digital display for live environmental monitoring.

Smart City with Automated Traffic Management – University Lab
Arduino

- Built an Arduino-based smart-city project with automated traffic management.

Face Match – Biometric Face Analysis *Python, OpenCV, DeepFace, MediaPipe, FastAPI* GitHub

- Built a computer-vision pipeline using RetinaFace alignment and FaceNet512 embeddings with cosine distance to determine whether two portraits show the same person.
- Extracted and visualised a 468-point facial mesh and normalised regional measurements for eyes, nose, mouth, jaw, face height, and eyebrows; exposed results through a FastAPI backend and React interface.

Local LLM Chatbot with RAG *Python, FastAPI, LangChain, ChromaDB, Docker* GitHub

- Developed a local LLM application with interchangeable Ollama and LM Studio providers, streamed responses, multimodal image/PDF attachments, and web-search augmentation.
- Implemented ingestion and retrieval for PDF, DOCX, text, and Markdown using LangChain chunking, provider embeddings, and persistent ChromaDB storage; packaged the services for Linux/Windows with Docker Compose.

Deep Reinforcement Learning for Intelligent Transportation Systems *Python, PyTorch, SUMO, DQN*

- Implemented and evaluated a DQN-based approach for dynamic route optimisation in urban traffic simulations.
- Analysed convergence and adaptability under changing traffic conditions, measuring congestion, fuel use, and emissions.

Reusable MFC Custom Controls Library *C++, MFC, GDI+, Visual Studio* GitHub

- Designed a modular library of 30 owner-drawn controls, including buttons, tabs, lists, sliders, status indicators, card panels, and a semiconductor wafer map.
- Separated reusable components from the demonstration application and documented APIs, integration patterns, resource dependencies, and known issues.

Data Analyzer *C++11, MFC, Visual Studio* GitHub

- Built a Windows engineering tool that parses structured text data, presents table and hierarchy views, and generates interactive charts with selectable axes.
- Organised file I/O, data vectors, tabbed analysis modules, chart controls, and image/data export into extensible application components.

Railway eService Platform & API *React, C#, ASP.NET Core 8, SQL Server* Web / API

- Developed a responsive ticketing platform and RESTful backend for route search, scheduling, bookings, fares, and user management with JWT authentication, password hashing, Swagger documentation, and SQL Server persistence.

SonicTheHedgeHog *C++, iGraphics/OpenGL* GitHub

- Built a 2D platformer with custom physics, frame-accurate collision handling, obstacle detection, and optimised rendering.

Achievements

- First place, Intra-AUST Programming Contest, Spring 2023 (100+ teams).
- First place, Intra-AUST Programming Contest, Fall 2022 (100+ teams).
- Competitive programming: Codeforces 1263 (Pupil), CodeChef 1542 (3 Star), and 700+ problems solved across online judges.

Languages

Bengali: Native **English:** Fluent (IELTS 6.5) **German:** A1 (beginner)